Technical report 2

Part1: User stories:

1. **External stakeholders:**
    - **Hajj Abdelaziz Ramadan:**

      "As a **Farmers' representative**, I want the system to send farmers voice notes and not in texts, so the process can be easier for them and can receive the season updates easily."

      Acceptance criteria:

      Given there are seasonal updates when farmers receive messages then the messages should be voice notes and not in texts, so the process be easier for them

      INVEST evaluation:

INVEST	Ratings	Improvements
Independent	partially	It needs to focus on the data of the registered farmers and if there's any updates came to them
Negotiable	yes	No improvement needed
Valuable	yes	No improvement needed
Estimable	yes	No improvement needed
Small	yes	No improvement needed
Testable	yes	No improvement needed

      "As a **farmers' representative**, I want the system to communicate with them in Arabic language not in English, so the farmers who don't understand English can communicate easily."

      Acceptance criteria:

      Given there are any updates when the farmers receive the messages then the messages should be in Arabic not English so the messages could be understandable to them.

INVEST	Ratings	Improvements
Independent	Partially	It needs to focus on the data of the registered farmers and the farmers that are sent messages for updates
Negotiable	yes	No improvements needed
Valuable	yes	No improvements needed
Estimable	yes	No improvements needed
Small	yes	No improvements needed
Testable	yes	No improvements needed

2. **Executive stakeholder:**
    - **Nagwa Fathy:**

      "As a **financial manager**, I want the farmers get paid as soon as the buyer pays the money, so farmers can be paid correctly and on time"

      Acceptance criteria:

      Given that the payment process is successfully completed when the buyers pay for carbon credits then farmers get their money on time

      INVEST evaluation:

INVEST	Ratings	Improvements
Independent	Partially	Clarifying the payment process as this story depends on the buyer's user story
Negotiable	yes	No improvement needed
Valuable	yes	No improvement needed
Estimable	yes	No improvement needed
Small	yes	No improvement needed
Testable	yes	No improvement needed

"As a **financial manager**, I want the farmers to inform me with any wallet number updates through a phone call and to be informed to finance only, so wallet number could be protected, and the financial process goes safely."

Acceptance criteria:

Given there are changes happening in farmer's wallet numbers, when farmer confirms wallet numbers, then the finance department be informed by this change and the money be sent to him correctly.

INVEST evaluation:

INVEST	Ratings	Improvements
Independent	yes	No improvement needed
Negotiable	yes	No improvement needed
Valuable	yes	No improvement needed
Estimable	yes	No improvement needed
Small	partially	It needs more clarification for the responsibilities the system should do
Testable	yes	No improvement needed

Part 2: Use Case Identification – Event decomposition technique:

I. state events identification:

Events	Entities identification	State change	Use case
Farmers register their plots	**plots**	As the registration goes from **unregistered** to **registered**	Registration of farmers' plots
Farms are grouped into Project Clusters of at least 500 hectares	**plots**	As farms go from **ungrouped** to **grouped** clusters	Grouping plots into large clusters
If a plot boundary overlaps with a plot already registered by someone else	**plots**	As it puts **registration on hold** till it's solved	Plot boundaries resolving
The satellite checks each plot for burning or land clearing	**plots**	As it **affects the verification process** and **counts flags** on every burning in the plot	Monitoring plots in each cluster
if plot flagged twice loses its eligibility for that season	**plots**	It prevents the plot from going through the verification process	Updating plot status
Farmers may also upload phone photos as evidence, but only photos with location and timestamps are accepted	**Photo evidence**	It changes flagged **plots** to **unflagged** or **only flagged once**	Uploading photo evidence
If more than 10 percent of the inspected plots fail, the whole cluster is rejected for that season	**clusters**	As if the required percentage failed, the cluster changes from **pending verification** to **rejected**, otherwise the cluster changes from **pending verification** to **accepted**	Verifying plots in each cluster

the registry issues credits with serial numbers	**Carbon credits**	The verified carbon credits change from **not issued** to **issued**	Issuing carbon credits for verified plots in each cluster
Credits are sold to companies at a fixed price	**Carbon credits**	The verified credits changes from **issued** to **sold**	Selling carbon credits
A credit that is used by a buyer must be retired	**Carbon credits**	The verified credits change from **sold** to **retired** so it can't be sold again	Retiring carbon credits
Farmers are paid by mobile wallet.	**Farmers' payment**	Changes from **not paid** to **paid**	Paying farmers
Revenue is split automatically between farmers, cooperative and the software team	**Farmers', cooperative's and team's shares**	Changes from **not paid** to **paid**	Paying farmers', cooperative's and team's shares

Perfect technology assumptions:

State events	Use cases	Perfect technology assumptions
Farmers register their plots	Registration of farmers' plots	**Keep, as its business registering activities**
If a plot boundary overlaps with a plot already registered by someone else	Plot boundaries resolving	**Keep, as its part of business registering activity**
Farms are grouped into Project Clusters of at least 500 hectares	Grouping plots into clusters	**Keep, as it part of the business activity**
The satellite checks each plot for burning or land clearing	Monitoring plots in each cluster	**Keep, as its business monitoring activity**
if plot flagged twice loses its eligibility for that season	Updating plot state	**Keep, as it's a part of business monitoring activity**
Farmers may also upload phone photos as evidence, but only photos with location and timestamps are accepted	Uploading photo evidence	**Keep, as its part of business monitoring activity**

If more than 10 percent of the inspected plots fail, the whole cluster is rejected for that season	Verifying plots in each cluster	**Keep, as its business verifying activity**
the registry issues credits with serial numbers	Issuing carbon credits for verified plots in each cluster	**Keep, as its business issuing credits activity**
Credits are sold to companies at a fixed price	Selling carbon credits	**Keep, as its business issuing credits activity**
A credit that is used by a buyer must be retired	Retiring carbon credits	**Keep, as its business issuing credits activity**
Farmers are paid by mobile wallet.	Paying farmers	**Keep, as its business payment activity**
Revenue is split automatically between farmers, cooperative and the software team	Paying farmers', cooperative's and team's shares	**Keep, as its business payment activity**

II. Event-Use case table:

Event Id	Event description	Event types	Triggering actor/ condition	Corresponding use case
E01	A field agent submits a new farmer's personal information, plot boundaries, and seasonal farming commitments.	**External**	**Field Agent**	Register Farmer and Plot
E02	A land officer submits a decision after investigating overlapping plot boundaries.	**External**	**Land Officer**	Resolve Plot Boundary Dispute
E03	A farmer submits geotagged and timestamped photographs as evidence of farming practices.	**External**	**Farmer**	Submitting Farming Evidence

E04	A verification body submits the results of its field inspections for a seasonal cluster.	**External**	**Verification Body**	Record Verification Results
E05	A verification body submits a signed verification report for an approved cluster.	**External**	**Verification Body**	Submit Verification Report
E06	A company requests to purchase available carbon credits at the fixed price.	**External**	**Company Buyer**	Purchase Carbon Credits
E07	A company submits a bid to participate in a quarterly carbon-credit auction.	**External**	**Company Buyer**	Submit Auction Bid
E08	The scheduled import satellite results are due	**Temporal**	**Every 10 days**	Import satellite results
E09	The scheduled carbon auction is due	**Temporal**	**At the end of every season**	Conduct credit auction
E10	The seasonal farmers' earning report is due	**Temporal**	**Every season**	Generate seasonal earning report
E11	Farmers register their plots	**State**	**plots**	Registration of farmers' plots
E12	If a plot boundary overlaps with a plot already registered by someone else	**State**	**plots**	Plot boundaries resolving
E13	Farms are grouped into Project Clusters	**state**	**plots**	Grouping plots into large clusters

	of at least 500 hectares			
E14	The satellite checks each plot for burning or land clearing	**State**	**plots**	Monitoring plots in each cluster
E15	if plot flagged twice loses its eligibility for that season	**State**	**plots**	Updating plot status
E16	Farmers may also upload phone photos as evidence, but only photos with location and timestamps are accepted	**State**	**Photo evidence**	Uploading photo evidence
E17	If more than 10 percent of the inspected plots fail, the whole cluster is rejected for that season	**State**	**clusters**	Verifying plots in each cluster
E18	the registry issues credits with serial numbers	**State**	**Carbon credits**	Issuing carbon credits for verified plots in each cluster
E19	Credits are sold to companies at a fixed price	**State**	**Carbon credits**	Selling carbon credits
E20	A credit that is used by a buyer must be retired	**State**	**Carbon credits**	Retiring carbon credits
E21	Farmers are paid by mobile wallet.	**State**	**Farmers' payment**	Paying farmers
E22	Revenue is split automatically between farmers, cooperative and the software team	**State**	**Farmers', cooperative's and team's shares**	Paying farmers', cooperative's and team's shares

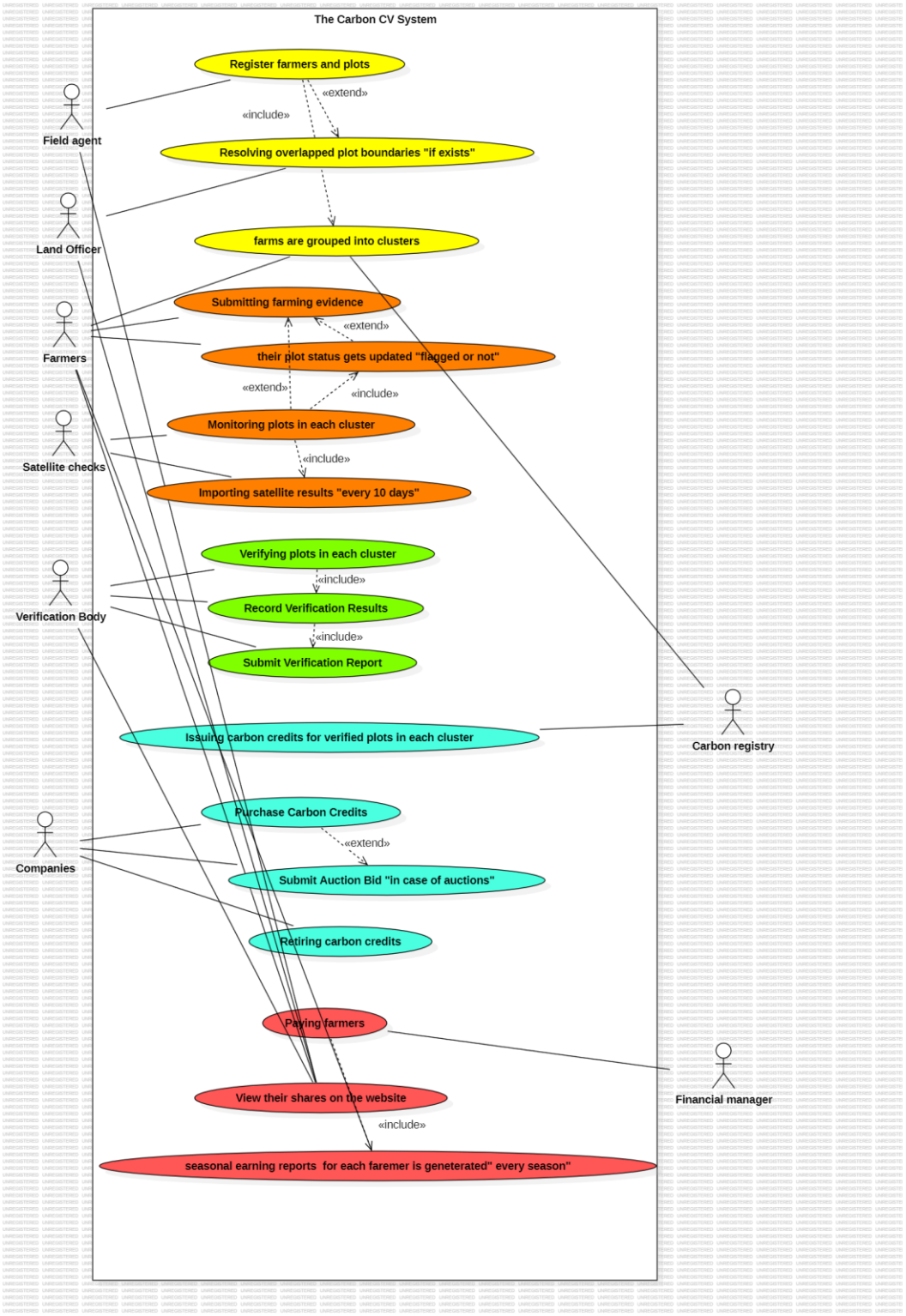
III. Brief use case description:

- **State events:**

Use case Id	Use case name	Brief use case description	Actors	Triggering events
E11	Registration of farmers' plots	Registers farmers' information on the system	**Field agents**	Farmers register their plots
E12	Plot boundaries resolving	Resolves any overlapping plots "if this problem exists"	**Land officer**	If a plot boundary overlaps with a plot already registered by someone else
E13	Grouping plots into large clusters	Group plots into clusters so it could reach at least 500 hectares	**None**	Farms are grouped into Project Clusters of at least 500 hectares
E14	Monitoring plots in each cluster	Monitors plots to check if there is any burning detected	**satellite**	The satellite checks each plot for burning or land clearing
E15	Updating plot status	Updates plots' status if there's any burnings happened and changes the flag	**None**	if plot flagged twice loses its eligibility for that season
E16	Uploading photo evidence	Updates plots and flag status as if farmers sent evidence photo, it would change the flag	**Farmers**	Farmers may also upload phone photos as evidence, but only photos with location and time stamps are accepted
E17	Verifying plots in each cluster	It checks a sample of plots failed to decide	**verifier**	If more than 10 percent of the inspected plots fail, the

		whether this cluster was rejected or accepted		whole cluster is rejected for that season
E18	Issuing carbon credits for verified plots in each cluster	It makes carbon credits issued to verified plots in the cluster	**Carbon registry**	the registry issues credits with serial numbers
E19	Selling carbon credits	Carbon credits status changes to sold after buyers purchase them	**Companies or buyers**	Credits are sold to companies at a fixed price
E20	Retiring carbon credits	Carbon credits status change to retired and can't be sold again after buyers purchase them	**Companies or buyers**	A credit that is used by a buyer must be retired
E21	Paying farmers	After the buyers pay for their purchased carbon credits, farmer's get their share through their wallets	**Financial manager**	Farmers are paid by mobile wallet.
E22	Paying farmers', cooperative's and team's shares	Everyone gets his share in the revenue after buyers purchase the carbon credits	**None**	Revenue is split automatically between farmers, cooperative and the software team

UML use cases diagram:



Part 3: Project Planning and Collaboration:

i. **Jira Project Timeline:**
https://students-team-lqungzw4.atlassian.net/jira/software/projects/KAN/boards/2?filter=&groupBy=none&atlOrigin=eyJpIjoiYTExMmY0Nzk4NmIzNDhjOGJmYjEyZDNjNWRlZjQ2NTYiLCJwIjoiaiJ9

ii. **GitHub repository:**
https://github.com/Rahaf-Osama/Carbon-CV-SWE-project-.git